Nilanjit Phulia

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.86
Bidhan Chandra Institution	Class XII (West Bengal Council of Higher Secondary Education)	2020	80.60%
St. Peter's School	Class X (Council for the Indian School Certificate Examinations)	2018	82.80%

SCHOLASTIC ACHIEVEMENTS

Prime Minister's Scholarship Scheme (PMSS) recipient

SKILLS

Programming: C, C++, Java, Python, SQL, MATLAB

Tools & Libraries: OpenCV, TensorFlow, Pandas, NumPy, Matplotlib, Seaborn, OpenPyXL, Scikit-learn

Domains: Data Analysis, Exploratory Data Analysis (EDA), Machine Learning, Computer Vision, Image Processing

RELEVANT COURSES

Machine Learning for Process Systems Engineering, Advanced Numerical Methods, SWOT Analysis and Risk Management

CERTIFICATIONS & AWARDS

Global Consumer Intelligence Course 2025 -- Matsuo-Iwasawa Laboratory, Graduate School of Engineering, The University of Tokyo

Microsoft Technology Associate (MTA): Introduction to Programming Using Python

Microsoft Applied Skills : NLP Solution with Azure AI Language

Microsoft Applied Skills : Streamline business workflows with AI chat

PROJECTS

India E-Waste Risk Analysis -- State-by-State Risk Assessment

Compiled a multi-source dataset integrating official Rajya Sabha parliamentary records (Sessions 263, 265, 266 -- data.gov.in) and UN Global E-Waste Monitor 2024 across 30 states/UTs and 7 years; performed end-to-end EDA including national trend analysis, state-wise YoY comparison, and Z-score outlier detection across **1.75M+** tonnes of annual e-waste data

Built a composite 100-point State Risk Scorecard across 3 dimensions -- collection volume, YoY growth, and reporting compliance -- categorizing 30 states into risk tiers; benchmarked India's recycling rate trajectory (9.8% to 43.5%) against **11** countries and delivered policy recommendations via 10 analytical visualizations

Contact Angle Detection of Droplets using Machine Learning

Developed an ML-based system to automatically detect droplet boundaries and compute contact angles across **1,000+** surface samples, improving measurement accuracy and eliminating subjectivity in wettability analysis

Implemented image processing techniques to analyze droplet profiles across different surfaces, enabling faster, repeatable and scalable evaluation for applications in coatings, material science, and surface engineering

Cooling Tower Fog Harvesting using Wettability Engineered Meshes

Analyzed **500+** mesh configurations using Python (Pandas, OpenCV) to quantify fog water capture; integrated calibration techniques converting pixel data into real-world measurements for accurate comparison

Identified key mesh parameters (pitch, fibre thickness, inclination) driving measurable improvements in water collection efficiency, enabling optimized mesh design for sustainable water recovery in industrial cooling towers

Number System Conversion

Developed a robust number system conversion tool supporting binary, decimal, octal, and hexadecimal formats, with accurate handling of both integer and fractional values across edge cases

Designed efficient algorithms and a user-friendly interface, enhancing computational accuracy while strengthening expertise in data parsing, error handling, and optimization for real-world applications

POSITIONS OF RESPONSIBILITY

Head Placement Coordinator, Greenko School of Sustainability (GSS)

Serving as primary liaison between OCS, Placement Coordinators, and GSS Batch of 2025-27 for end-to-end placement coordination

Coordinating with industry partners and prospective recruiters to organize placement drives and pre-placement talks

Class Representative (CR)

Served as a liaison between faculty and students for academic coordination; communicated important updates and resolved student concerns efficiently

Teaching Assistant, E-Waste Laboratory (EW6101)

Assisting in conducting laboratory sessions for junior students in the E-Waste Laboratory course, supporting hands-on learning in e-waste characterization/processing techniques.

Guided junior students through experimental procedures, grading, and lab report evaluation.